KG-1 BOOK A

Abacus Mental Arithmetic

Student Information

Name: .....

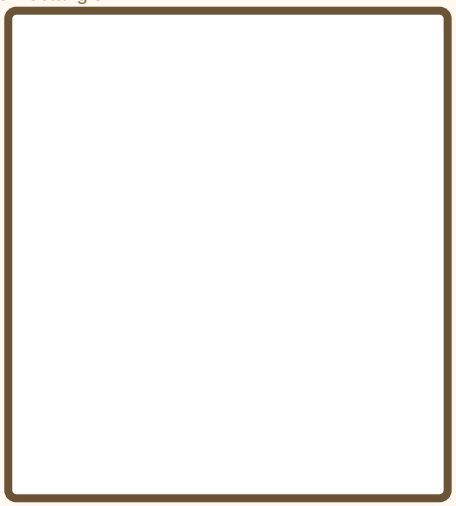
Level: .....

Date: .....

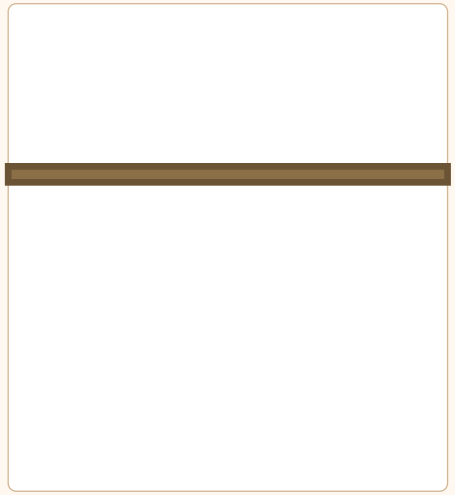
3 5 4

Description of the Abacus

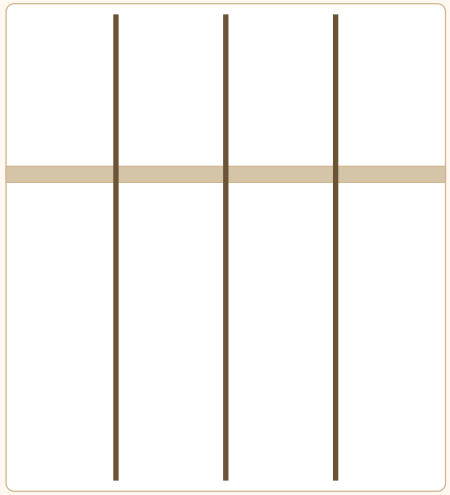
Frame
The outer rectangle



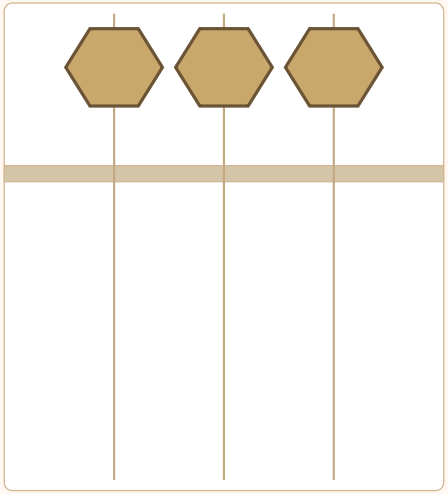
Beam
Horizontal divider bar



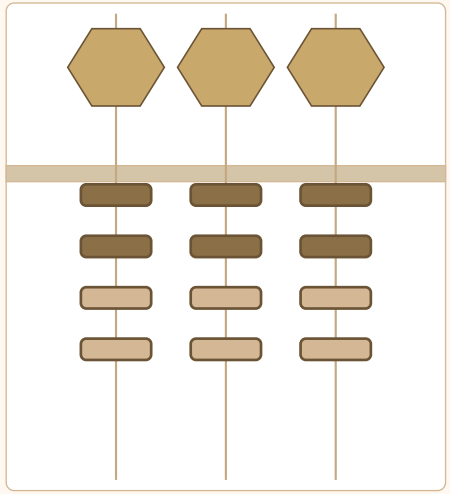
Rod
Vertical bars for beads



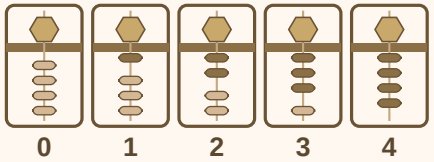
Upper Bead
Value = 5



Lower Bead
Value = 1 each



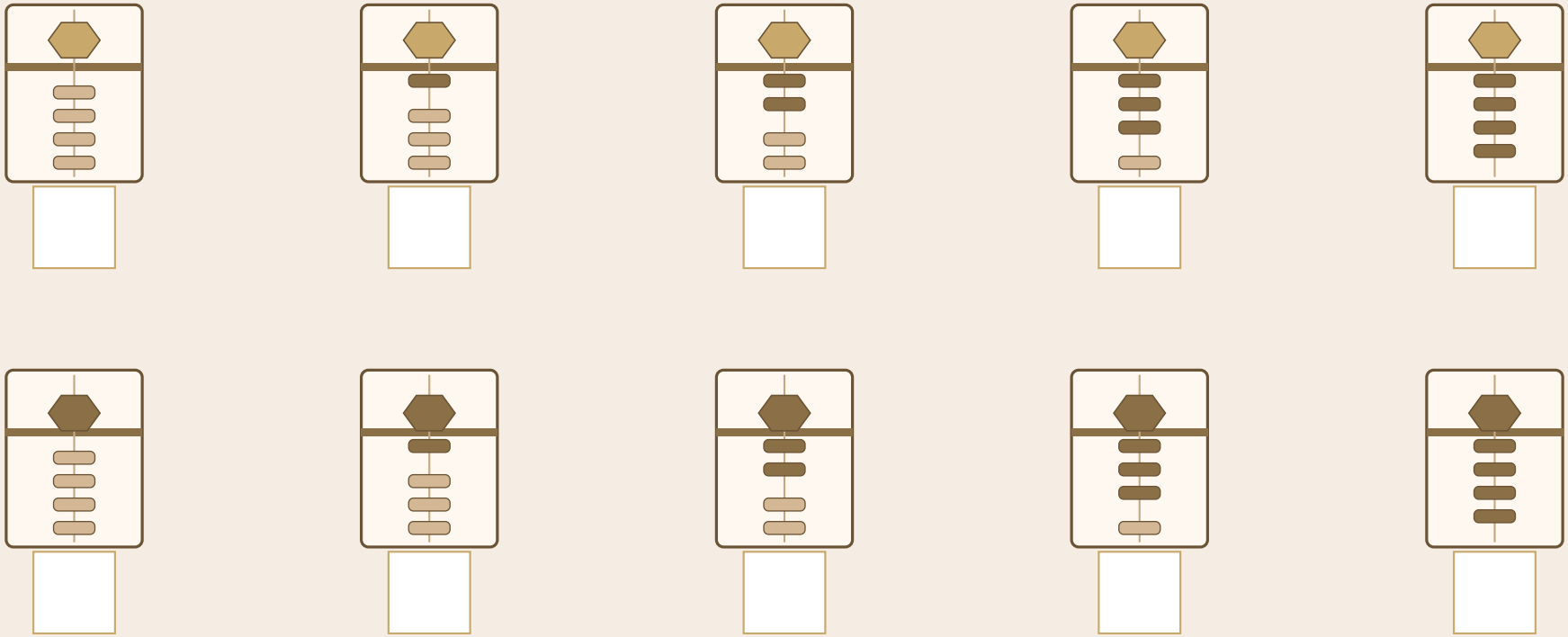
Complete Abacus
All parts together — values 0 to 9



Write the Value of Beads

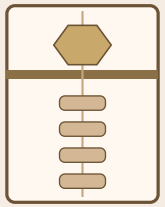
Set 1 to 9 on the abacus

Look at each abacus and write the number it shows in the box below.



Now set these numbers on the abacus:

3



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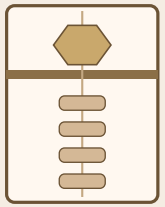
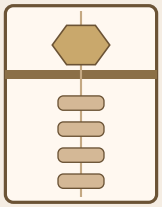
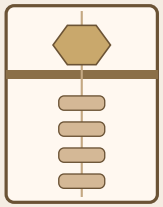


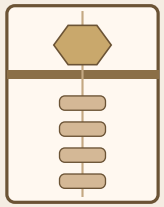
1



9

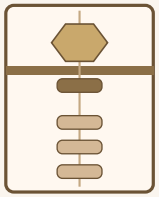
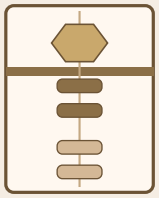
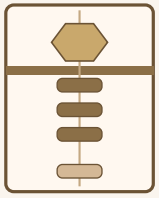
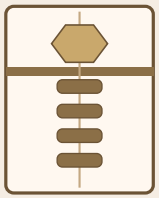
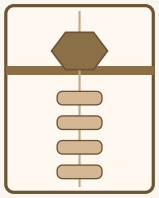


5



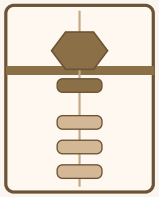
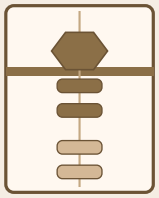
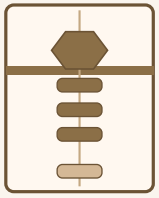
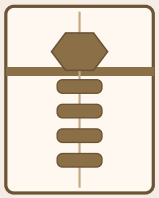
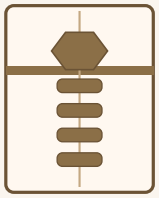
Understand Numbers

Numbers 1 to 5

Abacus	Hand	Dots	Number	Trace	Trace	Practice
		●	1	1	1	
		●●	2	2	2	
		●●●	3	3	3	
		●●●●	4	4	4	
		●●●●●	5	5	5	

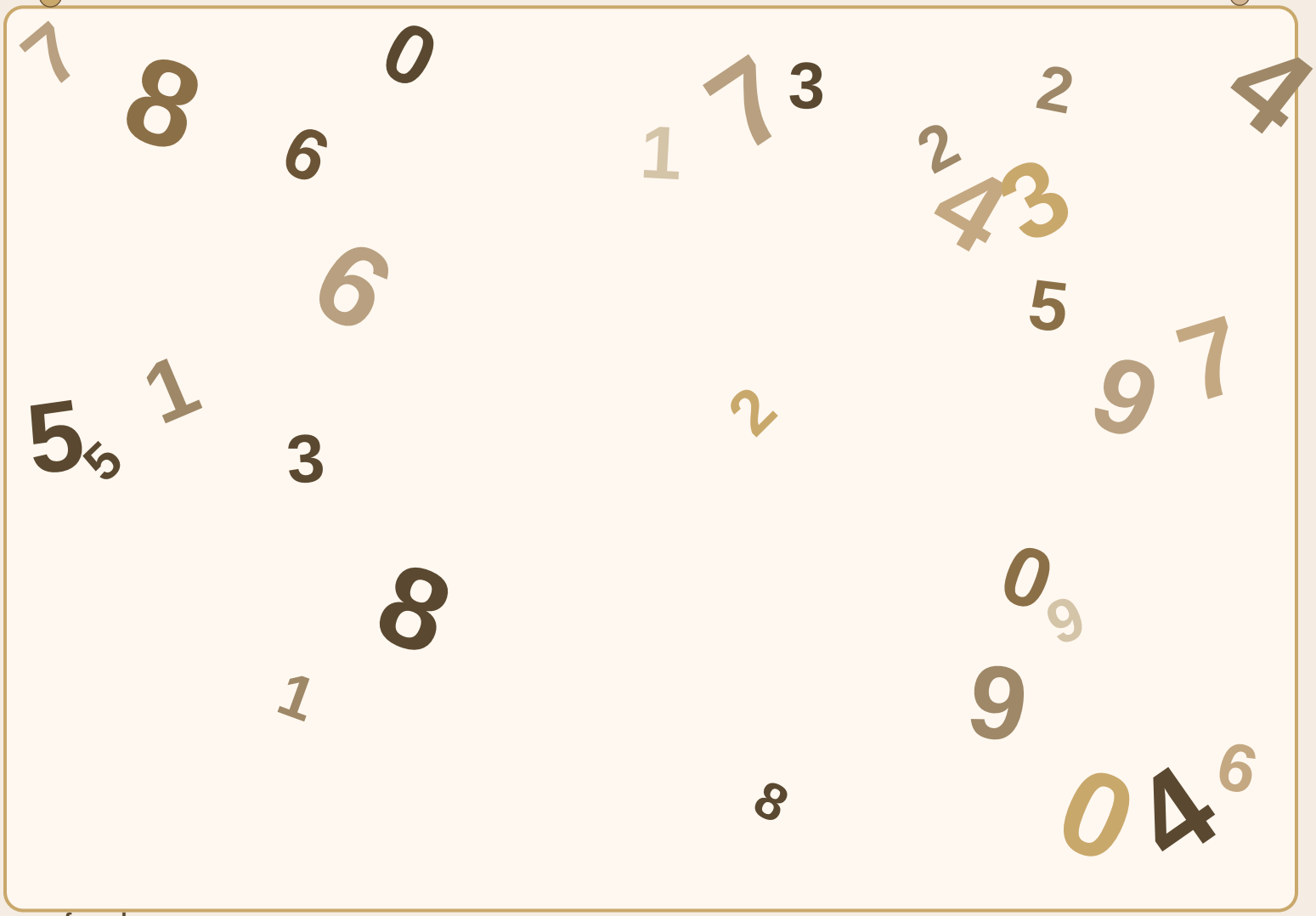
Understand Numbers

Numbers 6 to 10

Abacus	Hand	Dots	Number	Trace	Trace	Practice
	5+1		6	6	6	
	5+2		7	7	7	
	5+3		8	8	8	
	5+4		9	9	9	
	5+5		10	10	10	

Write the Numbers You Can Identify

How many numbers can you find hidden in the picture? Write them in the boxes!



Write the numbers you found:

Count and Write

Count the objects, write the number, then show it on the abacus.

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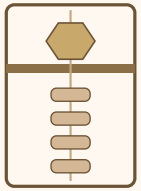

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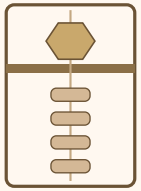

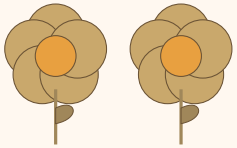

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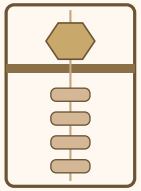

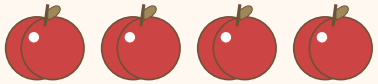

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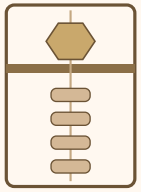

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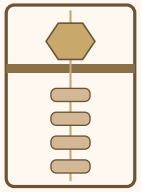

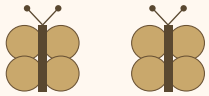

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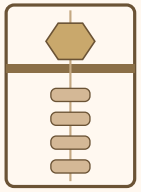
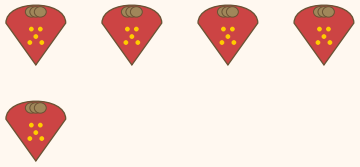


Count and Write

Count the objects, write the number, then show it on the abacus.



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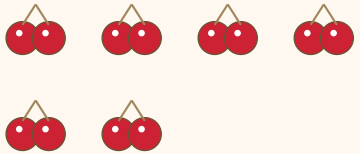

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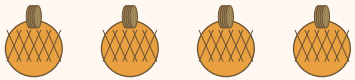

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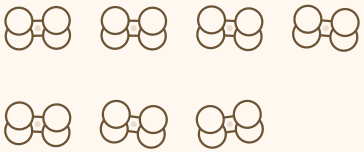

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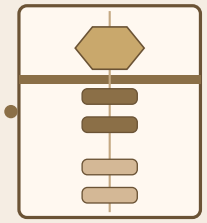
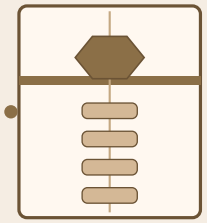
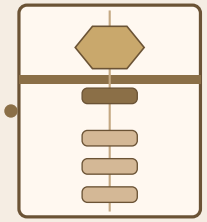
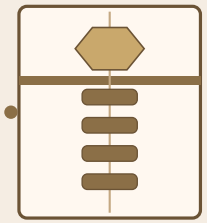
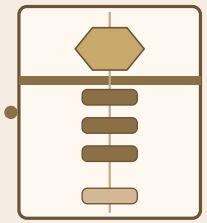
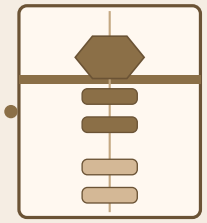
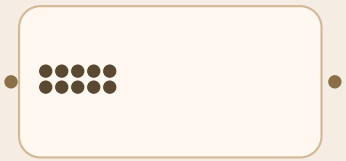
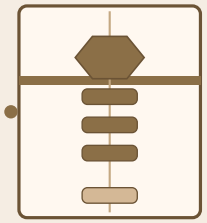
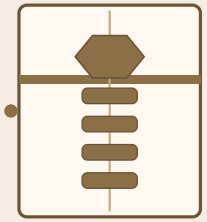
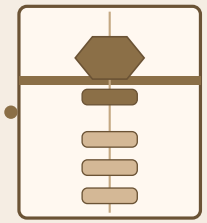
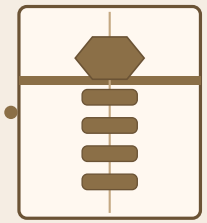
Match the Following

Draw lines to match each number with its dot pattern and abacus.

Number	Dots	Abacus
1	●●●●	
2	●●●●●	
3	●●●	
4	●	
5	●●	

Match the Following

Draw lines to match each number with its dot pattern and abacus.

Number	Dots	Abacus
6		
7		
8		
9		
10		

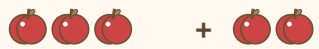
Abacus Calculation: 1 Digit 3 Rows

Fingering Ex: 1-1, 2-1, 3-1, 4-1, 2-2, 3-2

S.No.	1	2	3	4	5	6	7	8
1	1	2	3	4	1	3	4	2
2	+1	-1	-1	-2	+2	-2	-1	+1
3	-1	+1	-1	+1	-1	+2	-2	-2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	1	2	4	3	2	4	1	3
2	+3	-1	-3	-1	+2	-2	+4	+1
3	-2	+3	+1	+2	-3	-1	-3	-3
Ans.								

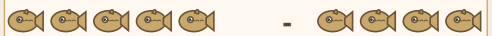
S.No.	1	2	3	4
1	4	2	3	1
2	-1	+1	-2	+2
3	-2	-3	+1	-3
Ans.				



3 + 2 =

5 - 3 =

4 + 1 =



5 - 4 =

Direct Addition & Subtraction

Fingering Ex: 1+2, 2+1, 3+1, 5+1, 2+2, 5+2

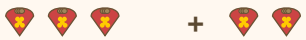
S.No.	1	2	3	4	5	6	7	8
1	1	2	3	4	1	2	3	4
2	+2	+1	+1	-2	+4	+3	+2	-3
3	-1	-2	-3	+1	-2	-4	-1	+2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	2	4	3	2	4	1	3	4
2	-1	-1	+1	+2	-2	+3	-1	-3
3	+3	-2	-2	-3	+1	-1	+2	+1
Ans.								

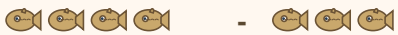
S.No.	1	2	3	4
1	1	3	2	4
2	+2	-2	+1	-1
3	+1	+2	-3	+1
Ans.				

2 + 4 =

5 - 2 =



3 + 2 =

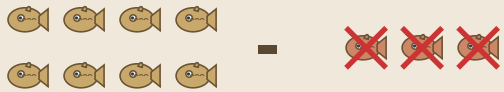


4 - 3 =

Direct Addition & Subtraction

Count the objects and write the answer.

5 + 3 =

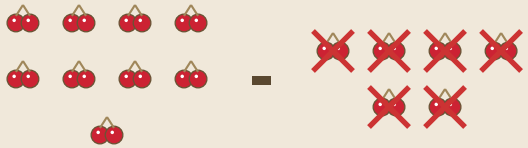


8 - 3 =

7 - 5 =

4 + 5 =

5 + 4 =



9 - 6 =

Abacus Calculation: 1 Digit 3 Rows

Fingering Ex: 1+5, 2+5, 3+5, 4+5, 5-5, 6-5

S.No.	1	2	3	4	5	6	7	8
1	1	2	3	4	1	2	3	4
2	+6	+5	+5	+5	+7	+6	+5	+5
3	-2	-3	-4	-6	-5	-1	-1	-2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	7	4	5	6	8	7	5	9
2	-5	-3	-2	-1	-5	-3	+3	-5
3	+3	+6	+4	-3	+1	-2	-6	-1
Ans.								

S.No.	1	2	3	4
1	1	7	3	8
2	+5	-2	+5	-5
3	-3	+1	-6	+2
Ans.				

1 + 5 =

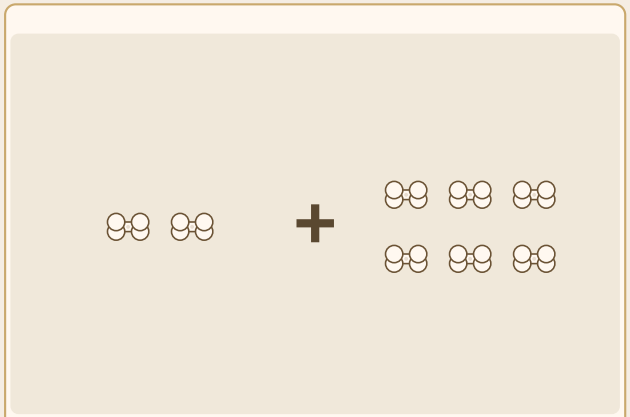
4 - 2 =

2 + 3 =

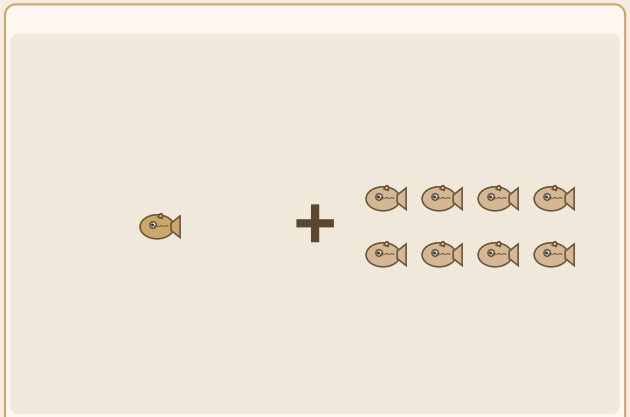
5 - 1 =

Direct Addition

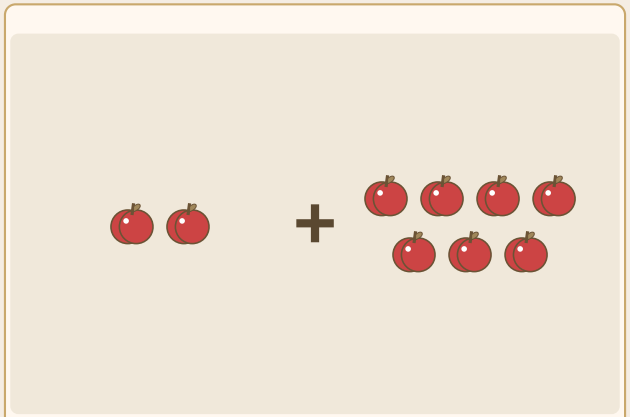
Count the objects and write the answer.



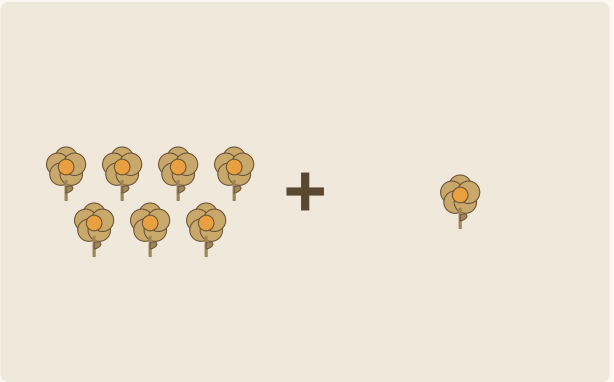
2 + 6 =



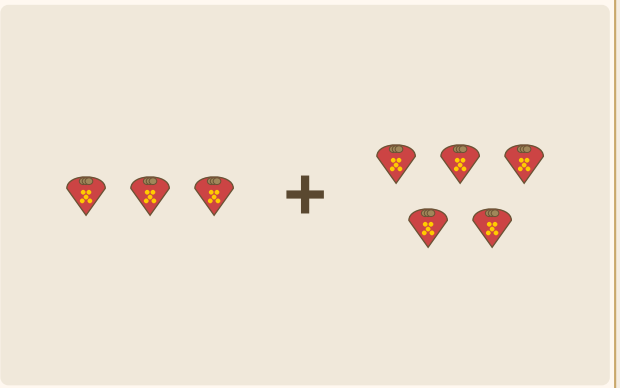
1 + 8 =



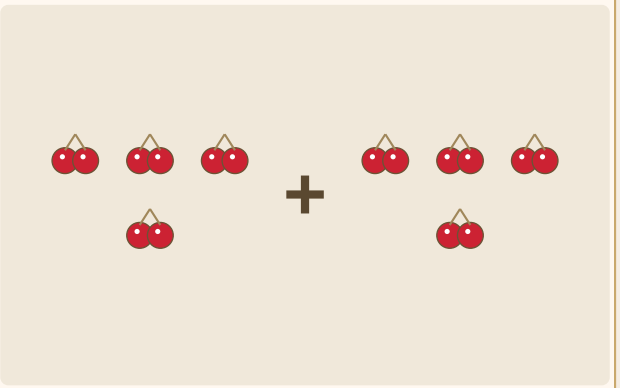
2 + 7 =



7 + 1 =



3 + 5 =



4 + 4 =

Abacus Calculation: 1 Digit 3 Rows

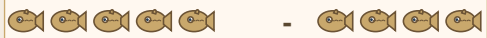
Fingering Ex: 1+6, 2+6, 3+6, 1+7, 2+7, 1+8

S.No.	1	2	3	4	5	6	7	8
1	1	2	3	4	1	5	6	7
2	+6	+7	-5	-2	+7	-3	-2	-5
3	-5	-2	+7	+5	-2	+1	-3	+1
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	1	2	3	4	8	7	6	9
2	+5	+8	-5	-3	-6	-5	-3	-1
3	+3	-6	+6	+5	+3	+4	+1	-8
Ans.								

S.No.	1	2	3	4
1	1	2	7	6
2	+7	+8	-5	-5
3	-5	-3	+1	+1
Ans.				

3 + 3 =



5 - 4 =

1 + 4 =

4 - 2 =

Abacus Calculation: 1 Digit 3 Rows

S.No.	1	2	3	4	5	6	7	8
1	3	7	5	2	1	4	6	8
2	+4	-2	-3	+7	+9	-2	-3	-5
3	-5	+4	+6	-5	-7	+5	-2	+3
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	4	9	2	5	7	8	3	1
2	+6	-4	-1	+3	-3	-6	+6	+8
3	-7	+1	+8	-6	+5	+4	-5	-7
Ans.								

S.No.	1	2	3	4
1	6	9	4	7
2	-2	-7	+5	-1
3	+1	+2	-3	-5
Ans.				

4 + 2 =

5 - 3 =

2 + 3 =

3 - 1 =

Abacus Calculation: 1 Digit 3 Rows

S.No.	1	2	3	4	5	6	7	8
1	4	8	6	3	7	1	5	2
2	-1	+1	-2	-3	-5	+8	+4	-1
3	-2	-7	+5	+9	+2	-6	-7	+6
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	8	6	3	9	4	7	2	5
2	-6	-1	+7	-2	+5	-3	+6	-4
3	+7	-3	-4	-5	-8	+1	-5	+3
Ans.								

S.No.	1	2	3	4
1	9	7	4	8
2	-2	-1	-3	-5
3	+1	-5	+5	-2
Ans.				

3 + 1 =

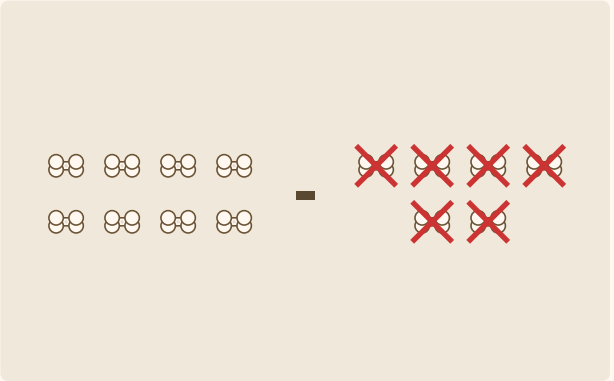
4 - 1 =

2 + 4 =

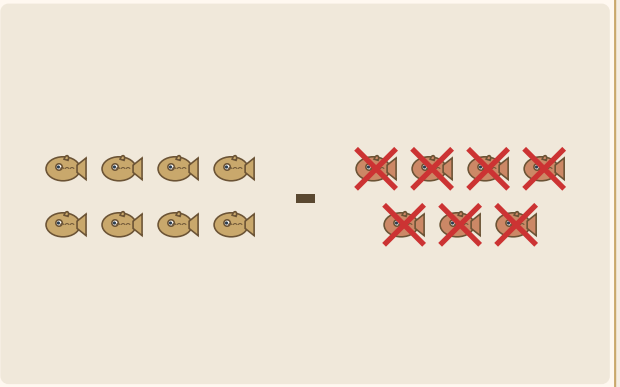
5 - 2 =

Direct Subtraction

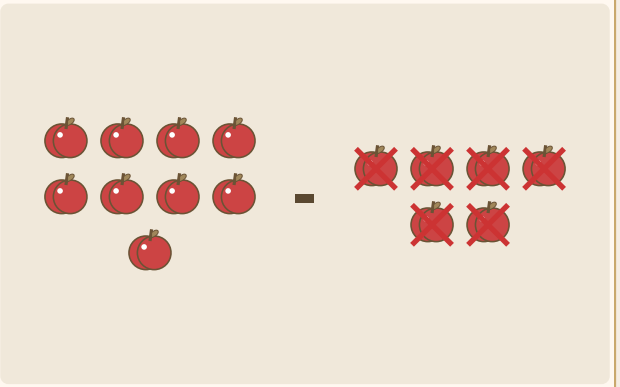
Count the objects and write the answer.



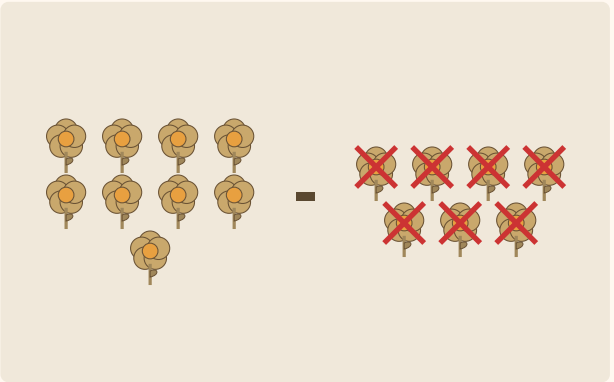
8 - 6 =



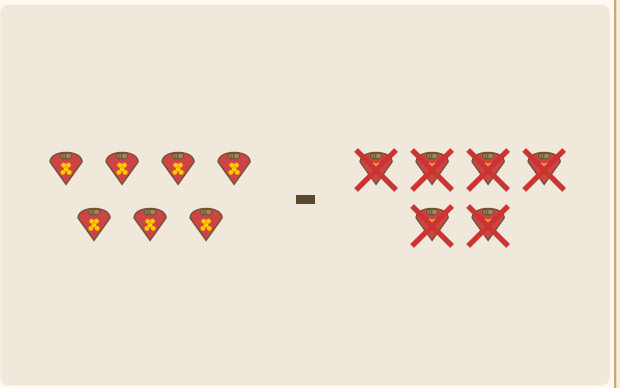
8 - 7 =



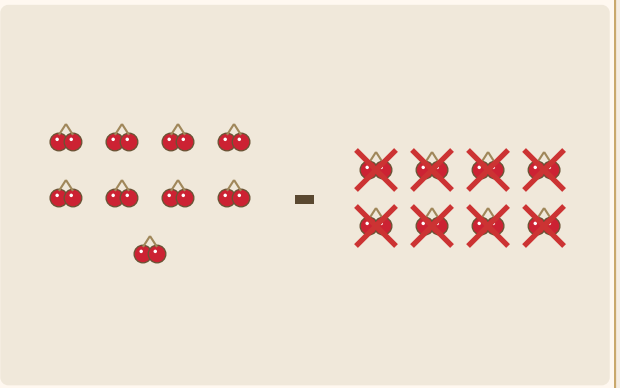
9 - 6 =



9 - 7 =



7 - 6 =



9 - 8 =

Abacus Calculation: 1 Digit 3 Rows

Fingering Ex: 6-6, 7-6, 8-6, 9-6

S.No.	1	2	3	4	5	6	7	8
1	7	8	9	6	5	4	3	2
2	-6	-6	-6	-6	+3	+6	-2	+7
3	+1	+3	-2	+7	-6	-8	+8	-6
Ans.								

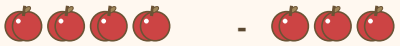
S.No.	1	2	3	4	5	6	7	8
1	6	7	8	9	6	5	4	3
2	+1	-3	-6	-6	+3	-4	+6	+7
3	-6	+5	+4	-1	-7	+8	-9	-6
Ans.								

S.No.	1	2	3	4
1	7	9	8	6
2	-6	-6	-6	-6
3	+6	+3	-1	+4
Ans.				

4 + 1 =

5 - 2 =

1 + 3 =



4 - 3 =

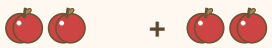
Abacus Calculation: 1 Digit 3 Rows

Fingering Ex: 7-7, 8-7, 9-7, 8-8, 9-8, 9-9

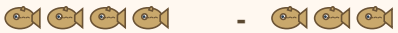
S.No.	1	2	3	4	5	6	7	8
1	7	8	9	8	9	9	7	8
2	-7	-7	-7	-8	-8	-9	-7	-8
3	+9	+4	-1	+5	+3	+8	+6	+9
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	9	7	8	9	8	9	7	8
2	-9	-7	-8	-7	-7	-9	-6	-7
3	+7	+8	+6	+4	+6	+5	+8	-1
Ans.								

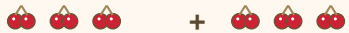
S.No.	1	2	3	4
1	9	7	8	9
2	-8	-7	-8	-9
3	+7	+9	+5	+8
Ans.				



2 + 2 =



4 - 3 =



3 + 3 =

5 - 4 =

Abacus Calculation: 1 Digit 3 Rows

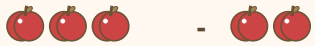
Fingering Ex: 8-8, 9-8

S.No.	1	2	3	4	5	6	7	8
1	1	7	8	4	9	2	6	3
2	+9	-8	-8	-8	-8	+8	-8	-8
3	-8	+9	+5	+9	+3	-9	+7	+8
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	9	7	8	5	3	1	4	9
2	-8	+2	-7	-8	+6	+8	-3	-8
3	+4	-8	+4	+9	-8	-7	+8	+6
Ans.								

S.No.	1	2	3	4
1	8	9	7	8
2	-8	-8	-8	-8
3	+7	+4	+9	+1
Ans.				

5 + 1 =



3 - 2 =

4 + 2 =

5 - 3 =

Abacus Calculation: 1 Digit 3 Rows

Fingering Ex: 8-8, 9-8, 9-9

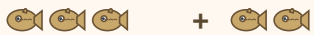
S.No.	1	2	3	4	5	6	7	8
1	9	8	7	9	8	7	9	8
2	-9	-8	-7	-8	-7	-6	-9	-8
3	+7	+9	+6	+4	+3	+5	+1	+2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	9	8	7	9	8	9	7	8
2	-7	-8	-6	-9	-7	-8	-6	-6
3	+4	+9	+8	+7	+6	+3	+5	-1
Ans.								

S.No.	1	2	3	4
1	9	8	7	9
2	-7	-8	-6	-9
3	+4	+9	+8	+7
Ans.				

1 + 4 =

5 - 1 =



3 + 2 =

4 - 2 =

Composition of 5

Understanding how numbers make 5

5 can be split into two parts in different ways!

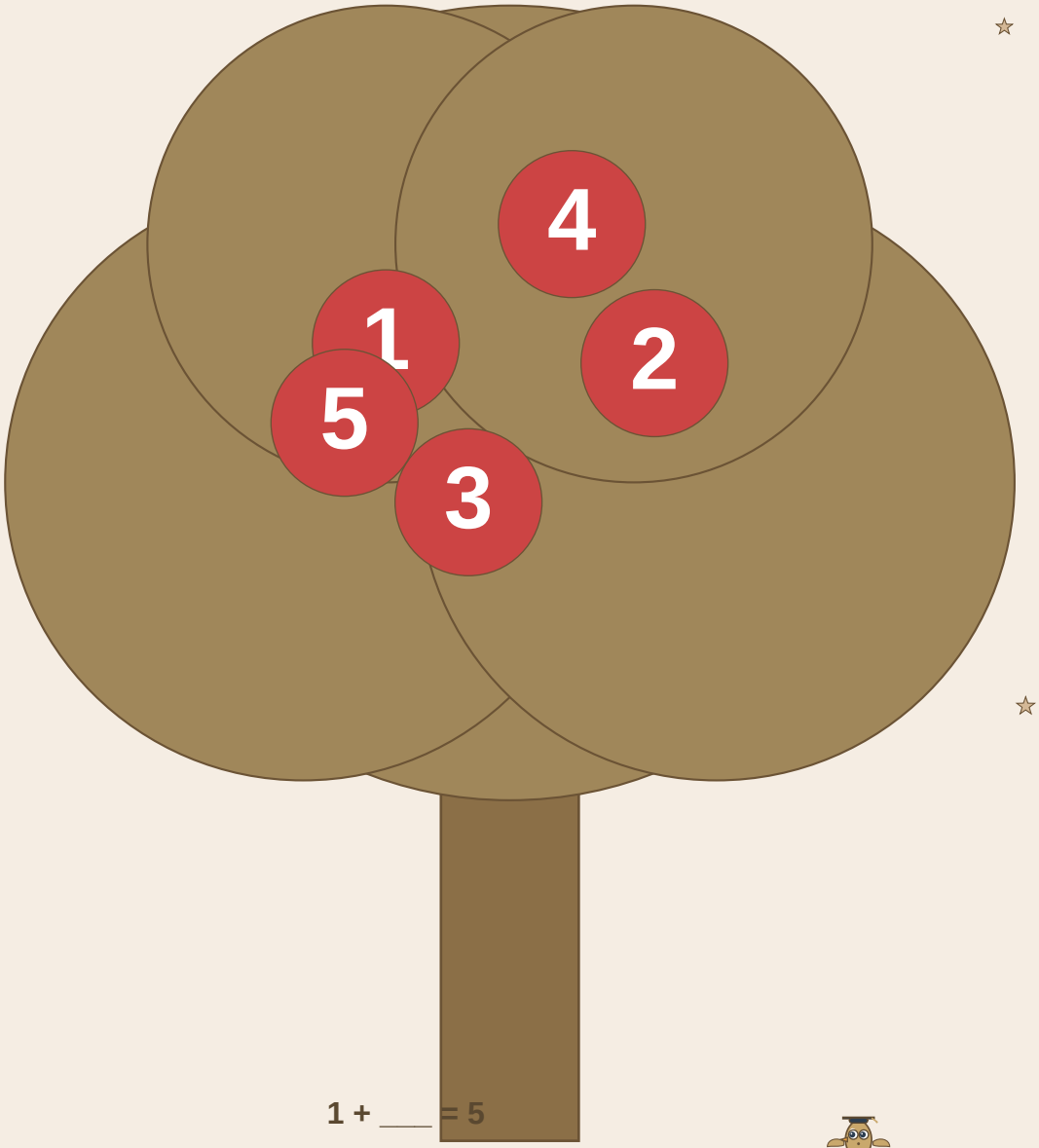
5

4 & 1 = 5

2 & 3 = 5

1 & 4 = 5

3 & 2 = 5



4 & 1 —— 5
2 & 3 —— 5
1 & 4 —— 5
3 & 2 —— 5

Fill in the missing number:

4 + ___ = 5

___ + 2 = 5

___ + 3 = 5

3 + ___ = 5

1 + ___ = 5

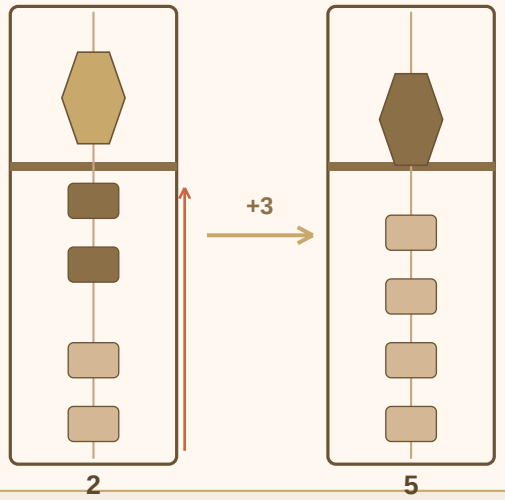
___ + 4 = 5

Composition of 5

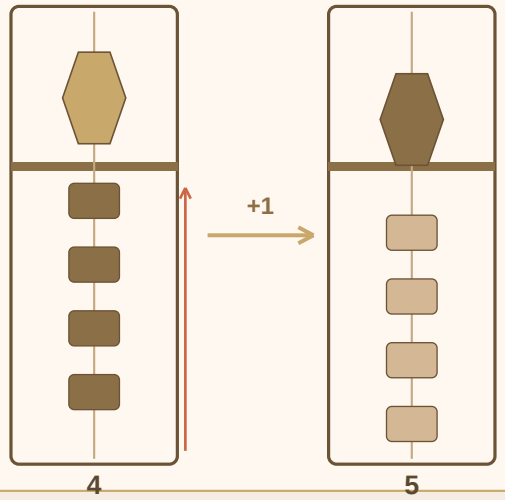
Showing combinations on the abacus

See how we make 5 on the abacus!

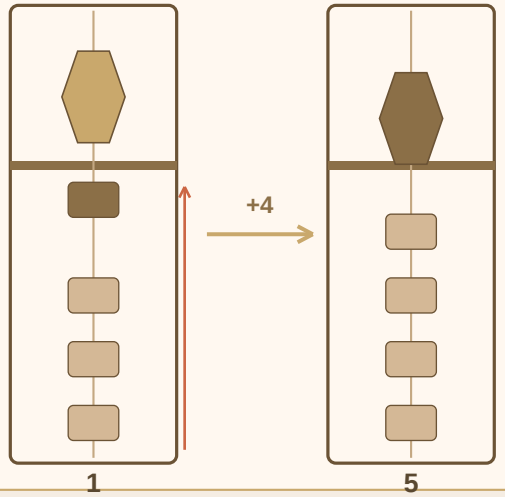
2 + 3 = 5



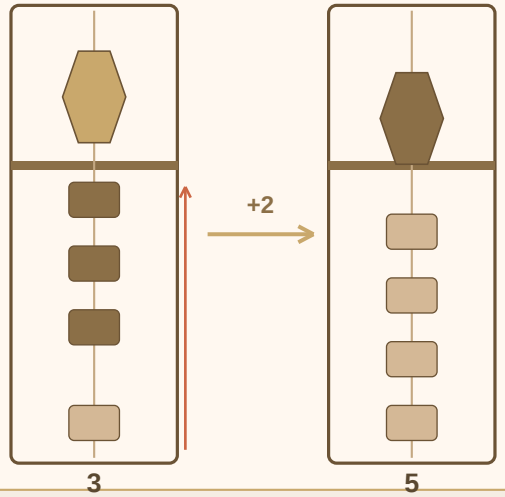
4 + 1 = 5



1 + 4 = 5



3 + 2 = 5



Abacus Calculation: Composition of 5 (+4)

+4=+5-1 | Fingering: 1+4, 2+4, 3+4, 4+4

S.No.	1	2	3	4	5	6	7	8
1	1	2	3	4	1	2	3	4
2	+4	+4	+4	+4	+4	+4	+4	+4
3	-3	-5	-6	-7	+2	-1	-5	-3
Ans.								

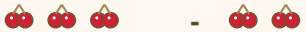
S.No.	1	2	3	4	5	6	7	8
1	2	1	3	4	1	2	3	4
2	+4	+4	+4	+4	+4	+4	+4	+4
3	-3	-5	+1	-6	-4	-5	-2	-1
Ans.								

S.No.	1	2	3	4
1	1	4	2	3
2	+4	+4	+4	+4
3	-3	-5	-6	+1
Ans.				

2 + 3 =

5 - 3 =

4 + 1 =



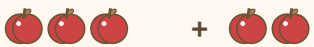
3 - 2 =

Abacus Calculation: Revision

S.No.	1	2	3	4	5	6	7	8
1	1	2	3	7	4	5	8	1
2	+3	+6	-1	-5	+2	-2	-6	-5
3	-2	-5	+4	+2	-3	+4	+3	+9
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	2	7	1	4	8	5	9	7
2	+4	-5	+3	-2	-3	+4	-6	+2
3	-3	+6	+5	+6	+1	-7	+2	-8
Ans.								

S.No.	1	2	3	4
1	3	2	6	5
2	+4	-1	+3	+2
3	-5	+7	-8	-4
Ans.				



3 + 2 =

4 - 1 =

2 + 3 =

5 - 2 =

Abacus Calculation: Composition of 5 (+3)

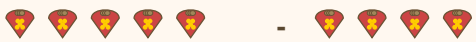
+3=+5-2 | Fingering: 2+3, 3+3, 4+3

S.No.	1	2	3	4	5	6	7	8
1	2	3	4	2	3	4	2	3
2	+3	+3	+3	+3	+3	+3	+3	+3
3	-4	-5	-6	+1	-1	-5	-3	-4
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	4	2	3	4	2	3	4	2
2	+3	+3	+3	+3	+3	+3	+3	+3
3	-2	-5	+1	-7	-4	-6	-3	-1
Ans.								

S.No.	1	2	3	4
1	3	4	2	3
2	+3	+3	+3	+3
3	-5	-4	-3	+2
Ans.				

4 + 2 =



5 - 4 =

2 + 4 =

4 - 1 =

Abacus Calculation: Revision

S.No.	1	2	3	4	5	6	7	8
1	3	2	4	7	5	8	1	6
2	+4	+6	-2	-3	+3	-5	+4	-4
3	-5	-3	+5	+1	-6	+2	-3	+3
Ans.								

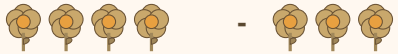
S.No.	1	2	3	4	5	6	7	8
1	2	4	5	7	1	8	3	9
2	+3	-3	-5	-2	+6	-3	+5	-7
3	-4	+6	+3	-4	-5	-4	-7	+2
Ans.								

S.No.	1	2	3	4
1	4	6	5	7
2	+3	-2	-3	-5
3	-5	+3	+2	+4
Ans.				

3 + 3 =

5 - 2 =

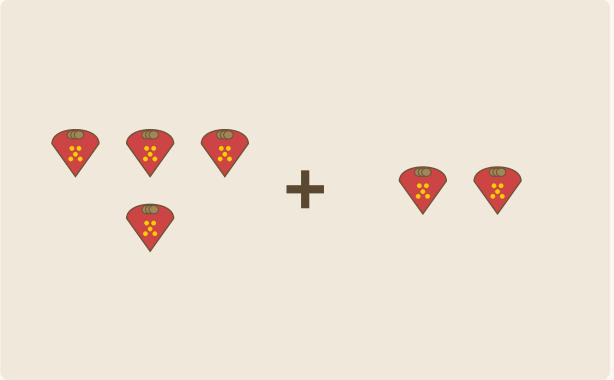
1 + 5 =



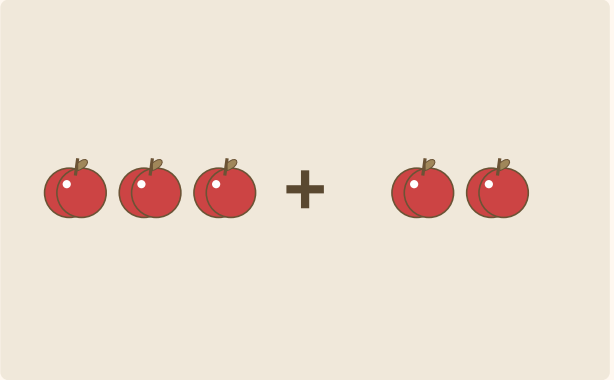
4 - 3 =

Count and Write

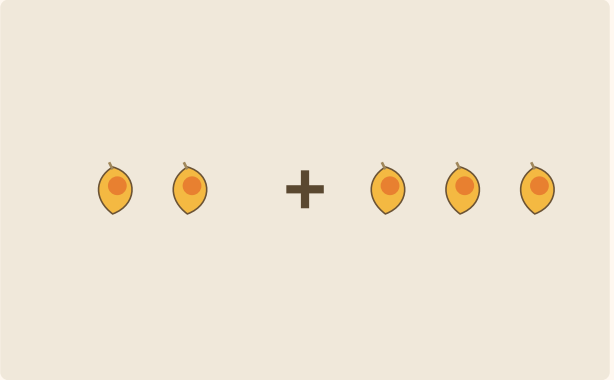
Count the objects on each side, then write the addition and answer.



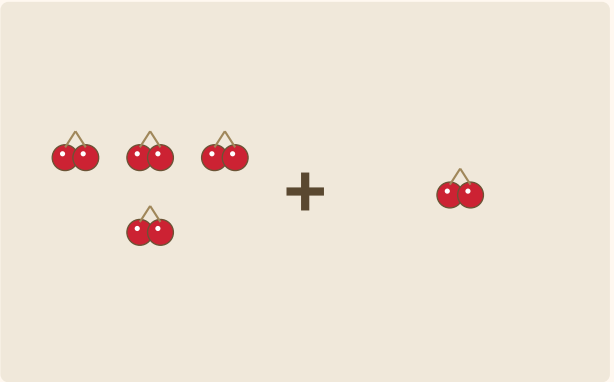
4 + 2 =



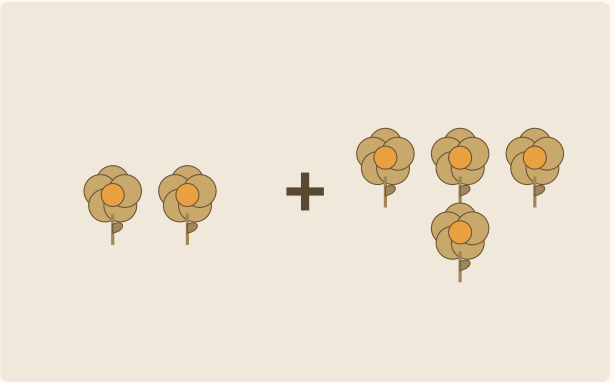
3 + 2 =



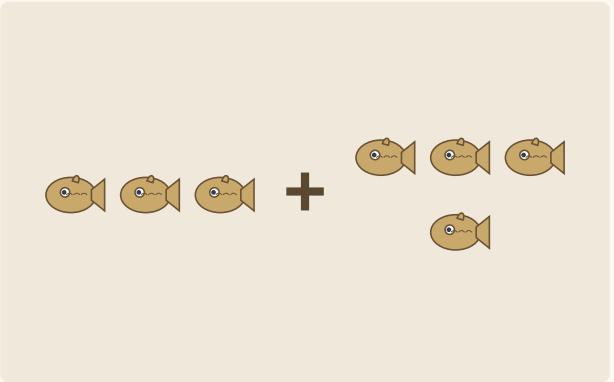
2 + 3 =



4 + 1 =



2 + 4 =



3 + 4 =

Abacus Calculation: Composition of 5 (+2)

+2=+5-3 | Fingering: 3+2, 4+2

S.No.	1	2	3	4	5	6	7	8
1	3	4	3	4	3	4	3	4
2	+2	+2	+2	+2	+2	+2	+2	+2
3	-4	-5	-3	-6	+1	-4	-2	-1
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	3	4	3	4	3	4	3	4
2	+2	+2	+2	+2	+2	+2	-2	+2
3	-5	-3	+2	-6	-4	-1	+4	-7
Ans.								

S.No.	1	2	3	4
1	3	4	3	4
2	+2	+2	+2	+2
3	-4	-6	-5	+1
Ans.				

2 + 2 =

4 - 2 =

3 + 1 =

5 - 3 =

Abacus Calculation: Revision

S.No.	1	2	3	4	5	6	7	8
1	2	4	5	7	3	6	1	8
2	+8	+3	+2	-3	+4	-5	+7	-4
3	-3	-5	-6	+1	-2	+4	-3	-2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	4	3	6	9	2	5	8	1
2	+2	+5	-4	-5	+7	-3	-2	+4
3	-3	-7	+2	-3	-1	+4	-5	+3
Ans.								

S.No.	1	2	3	4
1	4	7	5	3
2	+3	-5	+2	+4
3	-2	+3	-4	-6
Ans.				

5 + 1 =

3 - 1 =

2 + 3 =

4 - 2 =

Abacus Calculation: Composition of 5 (+1)

+1=+5-4 | Fingering: 4+1

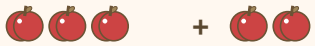
S.No.	1	2	3	4	5	6	7	8
1	4	4	4	4	4	4	4	4
2	+1	+1	+1	+1	+1	+1	+1	+1
3	-3	-5	+2	-4	-2	-6	+3	-1
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	4	4	4	4	4	4	4	4
2	+1	+1	+1	+1	+1	+1	+1	+1
3	-5	+6	-3	-5	-2	-4	+2	-5
Ans.								

S.No.	1	2	3	4
1	4	4	4	4
2	+1	+1	+1	+1
3	-3	-5	-4	+2
Ans.				

1 + 3 =

4 - 1 =



3 + 2 =

5 - 2 =

Abacus Calculation: Revision

S.No.	1	2	3	4	5	6	7	8	9	10
1	3	7	5	8	2	6	9	4	1	3
2	-4	-2	-5	-6	+4	-3	-7	+5	+7	+6
3	+6	+3	+4	+3	-5	+1	+3	-8	-3	-5
Ans.										

S.No.	1	2	3	4	5	6	7	8	9	10
1	5	8	6	3	4	7	2	9	1	5
2	+4	-3	-5	+7	-2	-6	+5	-4	+8	-2
3	-7	+2	+4	-9	+5	+8	-3	-3	-6	+4
Ans.										

S.No.	1	2	3	4	5	6	7	8	9	10
1	4	7	2	8	5	3	6	9	1	4
2	+6	-5	+4	-3	-4	+7	-2	-5	+3	+5
3	-8	+3	-3	+1	+6	-8	+4	-2	-4	-7
Ans.										

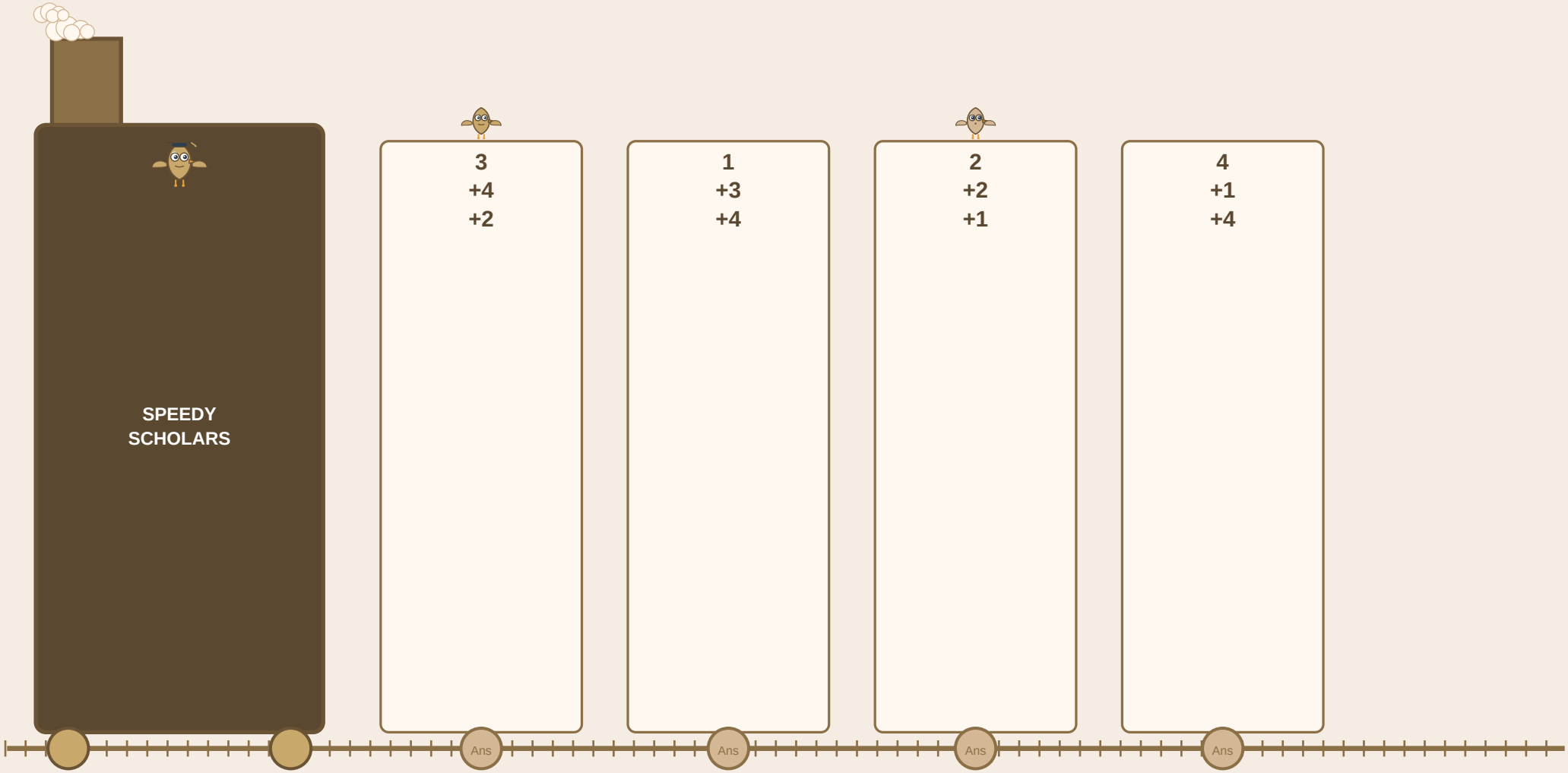
4 + 2 =

5 - 3 =

2 + 2 =

3 - 1 =

Abacus Calculation: Revision

All Aboard the Speedy Scholars Express!

Solve each problem — write answers in the wheel circles!